

Notes: Gruber (JF, 1996)
Another Puzzle: The Growth in Actively Managed Mutual Funds

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Institutional setting and motivation	3
4	Data and measurement	3
5	Models and empirical specifications	3
5.1	Raw return relative to the market	3
5.2	Single-index model	4
5.3	Four-index model	4
6	Identification and empirical design	4
7	Main results	5
7.1	Average active fund performance is negative	5
7.2	Index funds sharpen the puzzle	5
7.3	Closed-end funds show why pricing matters	5
7.4	Performance is persistent	5
7.5	Good funds do not simply charge away the benefit	5
7.6	Investors chase the right signals	5
7.7	The marginal dollar performs better than the average dollar	6
8	Tables and figures	6
8.1	Average performance and closed-end pricing	6
8.2	Performance persistence	7
8.3	Expenses and whether managers extract the rents	8
8.4	Investor flows and performance chasing	9
8.5	Return on new money	10
8.6	Persistence strategy versus index strategy	11
9	Short summary	11
10	Conclusion	12

1 Big picture

This paper asks a simple but important question: why do investors keep buying actively managed mutual funds when the average active fund underperforms passive benchmarks? The puzzle is especially sharp because mutual funds were one of the fastest growing financial intermediaries in the United States, yet the average actively managed equity fund did not beat a properly risk-adjusted benchmark.

The paper's answer is not that the average active fund is attractive. Instead, Gruber argues that open-end mutual funds are priced at net asset value, so managerial ability is not capitalized into the fund price. If skill exists and is not priced, then past performance can predict future performance, investors may rationally chase the right signals, and new money flowing into active funds may earn better returns than the average dollar already invested.

The paper therefore links four facts:

- active mutual funds underperform passive benchmarks on average;
- some measures of active fund performance are persistent;
- investor flows respond strongly to the same performance measures that predict future performance;
- the return earned on new cash flows into funds is better than the average fund return.

2 Incremental contribution of the paper

The incremental contribution is to reframe active mutual fund growth as a pricing and investor-selection puzzle rather than only a performance puzzle. Earlier work had shown weak average performance of mutual funds. Gruber adds that average performance is not the relevant object if investors can identify funds with predictably better performance before fees and if open-end fund shares are sold at net asset value.

The paper contributes in five concrete ways.

1. It evaluates active equity mutual funds using a multi-index benchmark that controls for market, size, growth/value, and bond exposures.
2. It emphasizes survivorship-bias correction through a “follow the money” approach when funds merge or disappear.
3. It compares open-end funds with closed-end funds to show why open-end pricing at NAV matters: management skill is not capitalized in open-end fund prices.
4. It shows that past performance predicts future performance, and that investor flows respond to the same information.
5. It asks whether the marginal dollar invested in active funds does better than the average dollar, creating a partial rational explanation for the growth of active management.

The main insight is subtle: even if active funds underperform on average, active funds can still attract rational investors if investors sort money toward predictably better managers and if the cost of acquiring that managerial ability is not embedded in the purchase price.

3 Institutional setting and motivation

Open-end mutual funds sell and redeem shares at net asset value. A fund with good management and a fund with poor management can both be purchased at the value of their underlying securities. This is different from closed-end funds, whose shares trade in the secondary market and can sell at a premium or discount to NAV. If investors expect a closed-end fund manager to be skilled, that expectation can be reflected in the market price of the fund.

This difference creates the central economic mechanism:

- In open-end funds, skill may be available at NAV and therefore may be underpriced.
- In closed-end funds, skill is more likely to be capitalized into the fund's market price.
- If skill or predictable performance exists, open-end fund investors may rationally chase performance.

This logic turns the puzzle into an empirical question: are investors chasing noise, or do their flows follow information that actually predicts future risk-adjusted returns?

4 Data and measurement

The main sample consists of common stock mutual funds listed in Wiesenberger's *Mutual Funds Panorama* at the end of 1984. The sample excludes foreign stock funds, specialized stock funds, balanced funds, and very small funds with less than \$15 million in assets. The initial sample contains 270 mutual funds, representing 77.2 percent of the assets held by all common stock funds at the end of 1984.

The study period is January 1985 through December 1994. Some funds merge or change policy during the period. To avoid survivorship bias, the paper follows investor money after fund disappearance rather than simply dropping nonsurviving funds.

Key objects:

- **Fund performance:** measured by raw returns, single-index alpha, and four-index alpha.
- **Fund expenses:** annual expense ratios, used to determine whether good funds charge higher fees.
- **Cash flows:** new money into and out of funds, measured both in dollars and normalized by beginning net asset value.
- **Closed-end fund price/NAV:** used to understand whether management ability is reflected in market prices when fund shares trade away from NAV.

5 Models and empirical specifications

The paper evaluates mutual fund performance with three related measures.

5.1 Raw return relative to the market

$$R_{it} - R_{mt}, \tag{1}$$

where R_{it} is the return on fund i and R_{mt} is the return on the S&P 500 index.

5.2 Single-index model

$$R_{it} - R_{ft} = \alpha_i^1 + \beta_{Mi}^1(R_{Mt} - R_{ft}) + e_{it}. \quad (2)$$

The intercept α_i^1 is the risk-adjusted excess return relative to the market factor.

5.3 Four-index model

$$R_{it} - R_{ft} = \alpha_i^4 + \beta_{Mi}^4(R_{Mt} - R_{ft}) + \beta_{Si}(R_{St} - R_{Lt}) + \beta_{Gi}(R_{Gt} - R_{Vt}) + \beta_{Bi}(R_{Bt} - R_{ft}) + e_{it}. \quad (3)$$

The four-index model controls for:

- the broad market factor;
- a small-minus-large stock factor;
- a growth-minus-value stock factor;
- a bond factor.

This specification is important because mutual funds are not random market portfolios. They tend to hold smaller and more growth-oriented stocks than the S&P 500. Without these controls, performance estimates can reflect style tilts rather than manager skill.

6 Identification and empirical design

The paper is not a causal identification paper in the modern quasi-experimental sense. Its empirical design is instead a sequence of linked tests that are meant to discipline the rational-investor interpretation.

1. **Average performance test:** compare active fund returns to benchmark indices.
2. **Index fund comparison:** ask whether low-cost index funds were realistic substitutes over the period.
3. **Closed-end comparison:** compare funds where management is priced to funds sold at NAV.
4. **Persistence test:** sort funds into deciles by past performance and examine future performance.
5. **Cash-flow test:** test whether investor flows respond to the performance measures that predict future performance.
6. **Return-on-new-money test:** compute whether the marginal dollar invested earns better risk-adjusted returns than the average dollar.

The logic is cumulative. Investors buying active funds look less puzzling if: (i) some funds are predictably better, (ii) expenses do not fully extract the value of performance persistence, and (iii) flows actually move toward better-performing funds.

7 Main results

7.1 Average active fund performance is negative

The average open-end equity mutual fund underperforms. Raw returns are below the market, single-index alphas are negative, and the four-index alpha is also negative. The paper reports that the average four-index alpha is about -0.054 percent per month, roughly -0.65 percent per year. This is smaller in magnitude than the raw return underperformance because the four-index model adjusts for the styles active funds actually hold.

Nonsurviving funds perform much worse than surviving funds, confirming the importance of avoiding survivorship bias. The nonsurviving-fund four-index alpha is about -0.229 percent per month.

7.2 Index funds sharpen the puzzle

By the mid-1990s, low-cost index funds existed and could track their targets very closely. The paper reports that low-cost index funds had annualized alphas close to their expense ratios, with tracking quality very high. Once low-cost index funds are available, the average underperformance of active funds becomes harder to explain by service or diversification benefits alone.

7.3 Closed-end funds show why pricing matters

Closed-end funds typically trade below NAV. The paper's Figure 1 shows the average market price/NAV ratio for nine closed-end funds generally below one during 1985-1994. Because closed-end fund shares trade in the market, investors appear to price the bundle of assets, fund structure, and management separately from the underlying NAV.

This contrasts with open-end funds, where investors buy and sell at NAV. If good management is not priced in an open-end fund, investors may rationally try to identify it using past performance.

7.4 Performance is persistent

Funds ranked high by past four-index alpha subsequently outperform low-ranked funds. Table II shows that the best-minus-worst decile spread in future four-index alpha is positive and statistically significant for both one-year and three-year selection windows. Table III shows similar persistence using raw monthly returns.

7.5 Good funds do not simply charge away the benefit

If managers with superior performance fully extracted their skill through higher expense ratios, performance persistence would not help investors. But Tables IV and V show that higher past performance does not correspond to meaningfully higher future expenses. In fact, worst past performers often have higher expense ratios than the best performers.

7.6 Investors chase the right signals

Table VI shows that funds ranked high by past four-index alpha receive large cash inflows, while low-ranked funds suffer outflows. Table VII shows that lagged four-index alpha predicts future cash

flows even after controlling for lagged cash flow and other return measures.

7.7 The marginal dollar performs better than the average dollar

Table VIII shows that the return on new cash flows is positive on a risk-adjusted basis. This is the core rationality result: investors' allocation decisions are not random. The average fund underperforms, but new money tends to flow into funds with better subsequent performance.

8 Tables and figures

8.1 Average performance and closed-end pricing

Table I
Average Monthly Performance

The table shows the average realized monthly percentage returns net of the S&P 500 and average coefficients from a time series regression of excess percentage returns against the single index and four index models. The sample consists of 270 open end funds and 9 closed end funds. The sample period is from January 1985 to December 1994.

Single Index Model:

$$R_i - R_f = \alpha_i^1 + B_M^1(R_M - R_f) + e_i$$

Four Index Model:

$$R_i - R_f = \alpha_i^4 + B_M^4(R_M - R_f) + B_S(R_S - R_f) + B_G(R_G - R_f) + B_B(R_B - R_f) + e_i$$

where

R_i = the return on fund i.
 R_f = the return on a thirty day T-Bill.
 R_M = the return on the S&P 500 index.
 $R_S - R_L$ = the difference in return between a small cap portfolio and a large cap portfolio.
 $R_G - R_V$ = the difference in return between a high growth portfolio and a high value portfolio.
 $R_B - R_f$ = the excess return on a bond index which represents an estimate of aggregate corporate and government bonds.

Fund Type	$R_i - R_M$	α^1	B_M^1	ρ^2	α^4	B_M^4	B_S	B_G	B_B	ρ^2
Open end funds										
All funds	-0.162	-0.130	0.963	0.810	-0.054	0.850	0.314	0.229	0.090	0.886
No load funds	-0.164	-0.127	0.964	0.800	-0.056	0.849	0.300	0.243	0.098	0.897
Load funds	-0.162	-0.134	0.966	0.830	-0.054	0.850	0.331	0.229	0.086	0.873
Maximum capital gains	-0.125	-0.182	1.073	0.780	-0.035	0.889	0.466	0.433	0.092	0.897
Growth	-0.154	-0.131	0.978	0.810	-0.045	0.860	0.306	0.264	0.077	0.886
Growth and income	-0.213	-0.073	0.829	0.840	-0.088	0.796	0.175	-0.031	0.108	0.878
Nonsurviving funds	-0.432	-0.350	0.929	0.710	-0.229	0.781	0.391	0.290	0.095	0.801
Closed end funds										
All funds, NAV return	-0.135	-0.031	0.864	0.780	0.014	0.787	0.193	0.153	0.077	0.820

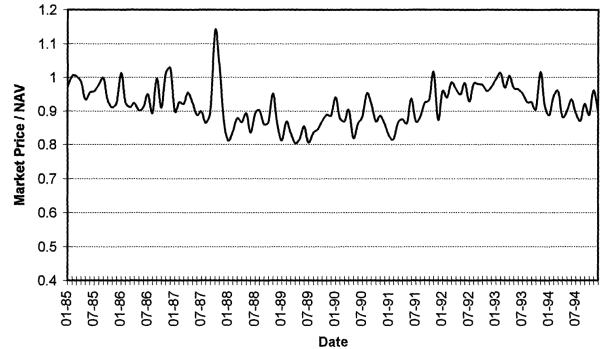


Figure 1. Average market price/net asset value. The figure shows the average market price/net asset value ratio for a sample of nine closed end funds from January 1985 to December 1994.

(a) Table I. Average monthly performance

(b) Figure 1. Average market price/net asset value

Read:

- Table I shows that open-end funds have negative average performance under all three measures. The four-index alpha for all funds is -0.054 percent per month.
- Nonsurviving funds perform especially poorly, with a four-index alpha of -0.229 percent per month.
- Closed-end funds have a positive NAV-based four-index alpha of 0.014 percent per month, but Figure 1 shows that their market prices generally trade below NAV.

Economic message: Active open-end funds underperform on average, but open-end pricing at NAV means that managerial ability may not be capitalized. Closed-end funds help illustrate the

contrast: when fund shares trade in markets, the bundle can trade at a discount to the assets.

8.2 Performance persistence

Table II

Performance Measured by Monthly Four Index Alpha

The table shows the average realized monthly four index alphas in percentage for one and three year holding periods where the deciles were formed using the ranking criteria shown at the top of each column. The sample consists of the 227 funds that existed in January 1982. The sample period is from January 1985 to December 1994.

Decile	Exp. Ratio ^a	1 yr return	1 yr 4 Index alpha	Exp. Ratio ^a	3 yr return	3 yr 4 Index alpha
	Panel A. 1 Year Holding Period			Panel B. 3 Year Holding Period		
Worst 1	-0.154	-0.200	-0.216	-0.143	-0.159	-0.223
2	-0.111	-0.033	-0.090	-0.079	-0.054	-0.044
3	-0.063	-0.022	-0.037	-0.069	0.023	-0.026
4	-0.018	-0.050	-0.044	-0.001	-0.036	-0.057
5	-0.017	-0.021	-0.020	-0.002	0.003	-0.013
6	0.023	-0.054	-0.039	0.010	0.013	-0.031
7	0.041	0.009	0.044	0.047	-0.013	0.011
8	-0.016	0.032	0.005	0.013	-0.049	0.005
9	-0.018	-0.016	-0.013	-0.018	-0.007	0.058
Best 10	-0.002	0.015	0.068	-0.008	0.028	0.073
Spearman Rank Coefficient	0.697**	0.782*	0.891*	0.552	0.503	0.903*
Top decile–bottom decile	0.152*	0.214**	0.284*	0.135*	0.187*	0.297*
Top decile–average of bottom 9	0.035	0.054	0.114*	0.018	0.059**	0.109*
Top 5 deciles–bottom 5 deciles	0.078*	0.062***	0.094*	0.088*	0.039**	0.096*

* Significant at 1 percent level.

** Significant at 5 percent level.

*** Significant at 10 percent level.

^a Expense ratios are reported from high to low.

Differences between deciles (row-folios) reported in the table is not as large as

(a) Table II. Performance measured by monthly four-index alpha

Table III

Performance Measured by Monthly Return

The table shows the average realized monthly percentage returns for one and three year holding periods where the deciles were formed using the ranking criteria shown at the top of each column. The sample consists of the 227 funds that existed in January 1982. The sample period is from January 1985 to December 1994.

Decile	Exp Ratio ^a	1 Year Return	1 Year 4 Index Alpha	Exp Ratio ^a	3 Year Return	3 Year 4 Index Alpha
	Panel A. 1 Year Holding Period			Panel B. 3 Year Holding Period		
Worst 1	0.977	0.936	0.879	1.003	0.944	0.874
2	0.995	1.080	1.066	1.035	1.086	1.083
3	1.073	1.112	1.060	1.056	1.164	1.093
4	1.096	1.064	1.103	1.122	1.095	1.081
5	1.092	1.104	1.098	1.123	1.135	1.110
6	1.155	1.092	1.065	1.124	1.116	1.137
7	1.120	1.146	1.184	1.147	1.117	1.142
8	1.126	1.162	1.175	1.148	1.104	1.132
9	1.124	1.087	1.072	1.124	1.113	1.189
Best 10	1.174	1.137	1.217	1.173	1.175	1.208
Spearman Rank Coefficient	0.903*	0.636	0.733**	0.927*	0.527	0.927*
Top decile–bottom decile	0.197**	0.201	0.338*	0.170*	0.231*	0.334*
Top decile–average of bottom 9	0.090***	0.050	0.139***	0.075**	0.077**	0.115**
Top 5 deciles–bottom 5 deciles	0.093*	0.066	0.101**	0.076*	0.040	0.113*

* Significant at 1 percent level.

** Significant at 5 percent level.

*** Significant at 10 percent level.

^a Expense ratios are reported from high to low.

(b) Table III. Performance measured by monthly return

Read:

- Table II sorts funds into deciles using past performance and shows that the best past performers continue to earn higher future four-index alpha.
- The best-minus-worst spread in future four-index alpha is 0.284 percent per month for the one-year holding period and 0.297 percent per month for the three-year holding-period analysis.
- Table III shows similar continuation using raw monthly returns, so the persistence result is not an artifact of one performance metric.

Economic message: There is performance predictability among active funds. This is the first necessary condition for a rational explanation of active-fund demand.

8.3 Expenses and whether managers extract the rents

Table IV

Annual Alpha and Annual Expenses in Performance Period when Ranking Four Index Alphas in Selection Period

The table shows the average annual four index alpha and expense ratio in percentage for a one year holding period where the deciles were formed using the one year four index alpha and for a three year holding period where the deciles were formed using the three year four index alpha. The sample consists of the 227 funds that existed in January 1982. The sample period is from January 1985 to December 1994.

Decile	1 Year Analysis		3 Year Analysis	
	α	Expenses	α	Expenses
Worst 1	-2.587	1.405	-2.679	1.539
2	-1.074	1.061	-0.524	1.044
3	-0.449	1.007	-0.310	1.034
4	-0.531	0.968	-0.679	1.035
5	-0.237	0.994	-0.159	0.924
6	-0.474	0.962	-0.366	0.976
7	0.531	0.986	0.134	0.949
8	0.062	1.016	0.055	0.996
9	-0.156	1.043	0.690	0.980
Best 10	0.819	1.052	0.879	1.029
SRC ^a	0.891*	0.164	0.903*	0.588

^a SRC = Spearman Rank Coefficient.

* Significant at 1 percent level.

(a) Table IV. Annual alpha and annual expenses

Table V

Expense Ratios for Deciles Formed on the Basis of Four Index Alphas

The table shows the annual expense ratios in percentage for deciles formed according to four index alpha. The expenses are shown for the period the deciles are formed and for each of the next four years. The sample consists of the 227 funds that existed in 1982. The sample period is from January 1985 to December 1994.

Decile(t)	t	Expenses			
		$t + 1$	$t + 2$	$t + 3$	$t + 4$
Worst 1	1.357	1.405	1.413	1.415	1.400
2	1.050	1.061	1.068	1.084	1.072
3	1.018	1.007	1.031	1.064	1.115
4	0.960	0.968	0.984	0.998	1.024
5	0.971	0.994	1.029	1.049	1.064
6	0.955	0.962	0.970	0.967	0.959
7	0.973	0.986	0.998	1.007	1.050
8	1.035	1.016	1.053	1.080	1.119
9	1.027	1.043	1.037	1.055	1.028
Best 10	1.042	1.052	1.048	1.058	1.057
Mean	1.039	1.049	1.063	1.078	1.089

(b) Table V. Expense ratios by performance decile

Read:

- Table IV shows that the worst past-alpha decile has strongly negative future annual alpha, while the best past-alpha decile has positive future alpha.
- Expense ratios do not rise monotonically with performance. The best decile does not charge enough more to eliminate the future performance advantage.
- Table V shows that future expense ratios remain close across most deciles, and the worst decile often has the highest expenses.

Economic message: Better past performers do not simply charge away their advantage. This supports the idea that investors can obtain some of the benefit of predictable performance by reallocating money across funds.

8.4 Investor flows and performance chasing

Table VI

Annual Cash Flows for Deciles Formed on the Basis of Four Index Alphas in the Year Following Formation

The table shows the average realized annual cash flow in millions of dollars and cash flow normalized by dividing by total net assets from the end of the previous year for deciles formed on the basis of one year four index alpha. The cash flows are for the year following decile formation. The sample consists of the 227 funds that existed in January 1982. The sample period is January 1985 to December 1994.

Decile	Cash Flow	Normalized Cash Flow
Worst 1	-40.35	-0.154
2	-42.74	-0.112
3	-4.16	-0.051
4	14.32	-0.050
5	7.93	-0.037
6	20.79	-0.024
7	48.51	0.045
8	75.77	0.066
9	97.59	0.128
Best 10	152.44	0.290
Spearman Rank Coefficient	0.98*	1.00*
Top decile-bottom decile	192.79*	0.444*
Top decile-average of bottom nine	132.70*	0.311*
Top five deciles-bottom five deciles	92.02*	0.183*

* Significant at 1 percent level.
** Significant at 5 percent level.

(a) Table VI. Annual cash flows by four-index-alpha deciles

Table VII

Prediction of Cash Flows

The table shows the coefficients from a regression of the realized cash flows against the four index alpha from the prior two years, the cash flow from the prior year, the one index alpha from the prior year, and the return from the prior year. The sample consists of the 227 funds that existed in January 1982. The sample period is from January 1985 to December 1994.

$$CF_t = \gamma_0 + \gamma_1\alpha_{t-1}^4 + \gamma_2\alpha_{t-2}^4 + \gamma_3CF_{t-1} + \gamma_4\alpha_{t-1}^1 + \gamma_5R_{t-1} + e_t$$

γ_0	γ_1	γ_2	γ_3	γ_4	γ_5	ρ^2
0.018*	0.235*					0.106
0.014*	0.222*	0.112*				0.135
0.004	0.184*	0.054*	0.364*			0.244
0.015*	0.111**	0.059*	0.345*	0.081*		0.258
0.014*	0.127**	0.062*	0.355*		0.005**	0.258

* Significant at 1 percent level.
** Significant at 5 percent level.

(b) Table VII. Prediction of cash flows

Read:

- Table VI shows strong performance chasing. The worst decile loses about \$40 million in cash flow, while the best decile receives about \$152 million.
- Normalized cash flow moves from -0.154 in the worst decile to 0.290 in the best decile.
- Table VII shows that lagged four-index alpha predicts future cash flows, even when lagged cash flows and other return variables are included.

Economic message: Investors do not merely hold active funds passively. They move money toward funds with better past risk-adjusted performance, and the predictive variable investors follow is related to the variable that forecasts future performance.

8.5 Return on new money

Table VIII
Monthly Realized Cash Flow-Weighted Alpha

The table shows the monthly realized cash flow-weighted alpha in percentage computed by summing the product of the realized four index alpha and realized cash flows for holding periods of one quarter, one year, and three years. Results are shown for positive cash flows, negative cash flows, weighted average cash flows, and net cash flows as well as for realizations assuming the cash is invested at the end of the quarter (Panel A) and at the beginning of the quarter (Panel B) during which the cash flow is realized. Panel C shows realizations for load and no load funds assuming investment at the end of quarter. The sample consists of the 227 funds that existed in January 1982. The sample period is from January 1985 to December 1994.

	Holding Period			
Return on:	1 Quarter	1 Year	3 Year	
	Panel A: Buy at End of Quarter			
Positive cash flows	0.0293	0.0241	0.0033	
Negative cash flows	0.0712	0.0187	0.0082	
Weighted average	0.0440	0.0221	0.0055	
Net cash flow	0.1470	0.0824	0.0492	
	Panel B: Buy at Beginning of Quarter			
Positive cash flows	0.1032	0.0466	0.0128	
Negative cash flows	0.0522	0.0214	0.0139	
Weighted average	0.0852	0.0376	0.0133	
Net cash flow	0.2885	0.1318	0.0813	
	Panel C: Load vs. No Load			
	1 Quarter		1 Year	
	No Load	Load	No Load	Load
Positive cash flows	0.0433	0.0199	0.0183	0.0281
Negative cash flows	0.0520	0.0834	0.0139	0.0217
Weighted average	0.0463	0.0425	0.0168	0.0257
Net cash flow	0.1456	0.1481	0.0584	0.1003

Table VIII. Monthly realized cash-flow-weighted alpha

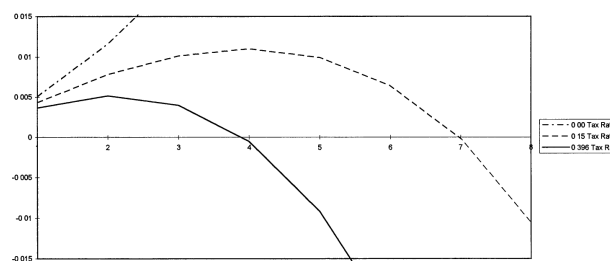
Table VIII. Monthly realized cash-flow-weighted alpha

Read:

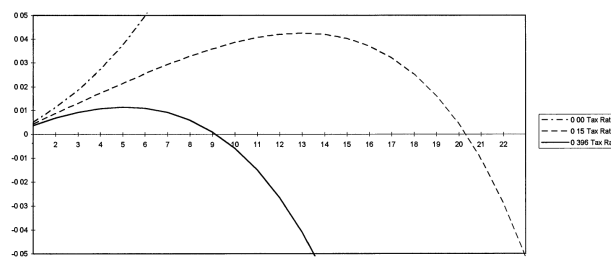
- Table VIII measures performance from the perspective of new cash flows rather than the average dollar already invested.
- For positive cash flows invested at the end of the quarter, the one-year cash-flow-weighted alpha is 0.0241 percent per month.
- When money is assumed to be invested at the beginning of the quarter, the one-year alpha for positive cash flows is 0.0466 percent per month.

Economic message: The marginal investment dollar does better than the average dollar in active funds. This is the strongest piece of evidence that investors' active-fund allocations may be more rational than average fund performance suggests.

8.6 Persistence strategy versus index strategy



(a) Figure 2. Net advantage of persistence strategy, 1985-1994 market environment



(b) Figure 3. Net advantage of persistence strategy, 1926-1994 market environment

Read:

- These figures compare an active-fund persistence strategy with an index strategy under different tax rates and holding periods.
- The advantage of the persistence strategy depends strongly on taxes and the holding horizon.
- Higher tax rates reduce the attractiveness of trading across active funds because reallocating between funds realizes taxable gains.

Economic message: Even if investors can identify better active funds, taxes and turnover costs matter. The rational case for active management is strongest when investors can exploit persistence without losing too much to taxes and expenses.

9 Short summary

The paper's core argument is that the growth of actively managed funds is not fully irrational despite average underperformance. Active funds underperform index-based benchmarks on average, but open-end mutual funds allow investors to buy management at NAV. If some funds have predictable performance and if expenses do not fully absorb that predictability, investors can rationally chase past performance.

Empirically, the paper finds that:

- active equity funds underperform on average;
- nonsurviving funds perform much worse, so survivorship bias matters;
- low-cost index funds sharpen the puzzle, especially by the mid-1990s;
- past fund performance predicts future fund performance;
- good past performers do not simply charge higher fees that eliminate the gain;
- investors direct cash flows toward funds with better past performance;
- the marginal dollar invested earns better risk-adjusted performance than the average dollar.

10 Conclusion

Gruber's paper gives a nuanced answer to the active-management puzzle. The average investor in the average active fund loses relative to an index benchmark. But the marginal investor may not be average: cash flows move toward funds with better past risk-adjusted performance, and new money earns positive cash-flow-weighted alpha.

The paper therefore separates two questions that are often conflated. First, do active mutual funds beat index funds on average? The answer is no. Second, can investors use available information to allocate toward better active funds? The paper's evidence suggests yes, at least to some extent.

The main limitation is that the evidence is predictive rather than causal. Performance persistence could reflect skill, fund style not fully captured by the model, time-varying risk, or other frictions. The paper also predates the explosive growth of very low-cost passive funds and modern ETFs. Still, its central contribution remains important: open-end mutual fund pricing at NAV creates a setting where managerial ability may not be priced directly, making performance predictability and flow allocation central to understanding why active funds survive and grow.